

Huntian Core Computing: A Three-Dimensional Macroscale Optoelectronic Hybrid

Brain-Inspired Computing Architecture Based on Icosahedral Modular Integration

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Abstract

Inspired by three sources — the armillary sphere model of the universe in traditional Chinese culture, the multi-layered nested concentric hollow structure of demon-work balls (ivory puzzle balls), and the engineering principles of the fourth-generation nuclear energy system, the pebble-bed high-temperature gas-cooled reactor (fuel pebble random close packing, continuous refueling, and circulation cooling) — this paper proposes a novel computing architecture named "Huntian Core Computing" (HCC). Centered on the concept of the "Computing Pebble Core" (CPC), this architecture fundamentally abandons the planar chip paradigm that has dominated the semiconductor industry for over half a century, adopting instead a macroscopic three-dimensional sphere based on modular assembly of regular icosahedral facets as the computing substrate.

In response to the three physical red lines faced by post-Moore planar chips — the planar interconnect bottleneck, the memory wall bottleneck, and the thermal density bottleneck — this architecture proposes four native integration designs: native integration of computing and memory to eliminate the memory wall, native integration of electronic computation and optical communication to break through the interconnect bottleneck, native integration of heat dissipation and structural framework to overcome the thermal density limit, and isomorphism between spatial topology and algorithmic models to achieve hardware-software co-optimization. Building on this foundation, this paper introduces for the first time a three-tier optoelectronic division-of-labor system to systematically reduce O/E conversion overhead, proposes a three-tier fault tolerance system to ensure decade-level system resilience, designs a hierarchical spherical routing architecture to exploit the natural parallelism and fault tolerance advantages of icosahedral topology, and establishes a revised quantitative performance model to address engineering feasibility concerns.

Topology selection analysis demonstrates that the regular icosahedron constitutes a Pareto-optimal solution across the three dimensions of manufacturing compatibility, spherical approximation

ratio, and mechanical stability: 20 congruent equilateral triangular facets require only a single set of photomasks for fabrication on standard 12-inch wafers, achieving a wafer area utilization of approximately 65%, a spherical approximation ratio of 0.94, and a maximum mechanical stress ratio of only 1.2. Through the systematic integration of cutting-edge technologies — including breakthroughs in monolithic heterogeneous integration of third-generation wide bandgap semiconductor materials such as silicon carbide (SiC) and gallium nitride (GaN), wafer-level icosahedral facet manufacturing and hybrid bonding assembly, embedded microchannel forced-convection cooling, 3D-stacked silicon photonic interconnects, and brain-inspired compute-in-memory based on spiking neural networks — this paper demonstrates the physical feasibility and technical accessibility of the "big first, strong first, precise later" engineering roadmap.

Revised quantitative models show that a 2035 engineering prototype (5 cm sphere diameter, integrating 2,000 nodes) is projected to achieve an energy efficiency ratio 6–8 times that of contemporary commercial GPUs, a computing density approximately 1/1,000 that of traditional server racks, communication latency as low as 10–15 nanoseconds, and a volume only approximately 1/18,000 that of a traditional server rack. Total cost of ownership analysis indicates that over a five-year lifecycle, an HCC cluster would achieve a TCO approximately 50% that of a conventional GPU cluster. A 2040 production model (2 cm sphere diameter, integrating 10,000 nodes) is projected to achieve a computing density approximately 1/100 that of traditional server racks.

This paper does not intend to provide a final engineering solution, but aspires through this paradigm proposal to offer a rigorous and bold reference answer — drawn from Eastern engineering wisdom — to the global semiconductor industry approaching physical limits, and to the fundamental post-Moore question of "what should computing look like?"

Keywords

Regular icosahedral modular integration; three-dimensional macroscale computing; optoelectronic hybrid computing; brain-inspired compute-in-memory; wide bandgap semiconductors; embedded microfluidic cooling; spherical routing algorithm; three-tier fault tolerance system

1 Introduction: From the "Planar Predicament" to the "Spherical Solution"

1.1 The Twilight of Moore's Law and Three Physical Red Lines

Since Gordon Moore proposed his famous eponymous law in 1965, the semiconductor industry has driven the digitalization of human civilization at the astonishing pace of doubling transistor density every 18 to 24 months. However, as manufacturing processes approach the sub-3-nanometer node, Moore's Law is encountering three insurmountable "red lines" from the physical world.

The First Red Line: The Planar Interconnect Bottleneck. In two-dimensional chips, signal traces are constrained by the physical limitations of Manhattan distance and resistance-capacitance (RC) delay. As line widths shrink, the resistance of metal interconnects rises sharply, parasitic capacitance increases, and signal delay along with transmission power consumption grow exponentially. According to projections from the International Technology Roadmap for Semiconductors (ITRS), when linewidth shrinks below 10 nm, interconnect delay will exceed transistor gate delay, becoming the primary factor limiting chip performance [1]. This is the fundamental physical reason why CPU clock speeds have stagnated at around 5 GHz for over a decade — it is not that the transistors themselves are not fast enough, but that the "roads" connecting them have become congested.

The Second Red Line: The Memory Wall Bottleneck. In the Von Neumann architecture, the central processing unit and memory are physically separated, requiring data to be shuttled back and forth repeatedly. According to Horowitz's classic energy analysis, in advanced process chips, reading a single 64-bit word from off-chip DRAM consumes approximately 200 times the energy of one double-precision floating-point operation [2]. In the field of AI accelerators, the energy consumed by data movement can account for over 80% of total computing energy consumption [3]. This is akin to a library and a reader separated by a highway, where the time spent reading is far less than the time spent commuting.

The Third Red Line: The Thermal Density Bottleneck. As transistor density increases, the heat flux density per unit area has surged to astonishing levels. Current high-performance graphics processing units (GPUs) have thermal design powers (TDP) exceeding 2,000 watts — NVIDIA's Rubin R100 GPU, announced at CES 2026, became the first production processor to break through this power limit [4]. The heat flux density of such chips is now comparable to that of a nuclear reactor core or a rocket engine nozzle, pushing conventional air-cooling and heat-sink solutions to their physical limits. It is estimated that, without a

fundamental change in cooling paradigm, the thermal density of high-performance chips after 2030 will exceed the heat flux density of the Sun's surface.

1.2 Eastern Wisdom on the Spatial Organization of Information Density

As the Western-dominated semiconductor industry struggles in planar competition, traditional Chinese culture's contemplation of information capacity and spatial organization has, since antiquity, exhibited a distinctly different dimensional orientation.

The Taiji Bagua (Eight Trigrams) constitutes a complete enumeration of a three-dimensional Boolean space through the ternary arrangement of Yin and Yang lines — one of humanity's earliest three-dimensional symbolic encoding systems. The armillary sphere, with its multi-layered concentric spherical structure simulating celestial coordinates, mapped the positional information of the sun, moon, and stars onto a three-dimensional spherical coordinate system, achieving a level of precision representing the pinnacle of ancient science [5]. The demon-work ball (ivory nested ball), with its exquisitely crafted structure of "hollowed layers, rings within rings, each layer rotating independently," demonstrated the extraordinary artistic imagination of the Chinese people for three-dimensional nested space: a sphere containing another sphere inside, which contains yet another — this is the most intuitive macroscopic metaphor for three-dimensionally integrated chips.

These three cultural heritages converge unanimously on a common principle: the optimal structure for information capacity, transmission, and circulation is not a two-dimensional planar grid, but a three-dimensional spherical unity. This principle provides a source of intellectual resources and design inspiration for chip technology trapped in the predicament of planar miniaturization, offering a way to break out of existing frameworks.

1.3 The Nuclear Reactor Core: An Engineering Paradigm Overlooked by the Computing Field

In the engineering domain, there exists another spherical structural paradigm that is far closer at hand yet has long been overlooked by the computing field — the core design of the pebble-bed high-temperature gas-cooled reactor.

The pebble-bed high-temperature gas-cooled reactor (PB-HTGR) is one of the preferred reactor types for fourth-generation nuclear energy systems, attracting widespread attention for its inherent safety and high outlet temperature [6]. Its core is formed by the random packing of hundreds of thousands of spherical fuel elements, each fuel pebble only 6 cm in diameter and 200 grams in weight, containing 7 grams of uranium — yet releasing energy equivalent to burning over 1.6 tons of standard coal. This is the ultimate embodiment of "small pebble, immense energy."

The operational logic of the HTGR forms a structural analogy with the computing architecture proposed in this paper across at least four dimensions:

First, the close-packed structure of spherical elements. Hundreds of thousands of fuel pebbles form the core through random close packing, naturally creating a highly dense three-dimensional arrangement under gravity. A connected pore network forms between the pebbles, which serves as the flow path for the helium coolant [7].

Second, through-flow convective cooling. The HTGR uses chemically extremely stable helium gas as the coolant, which flows through the pores between fuel pebbles, continuously carrying fission heat out of the core. This "through-flow" cooling mode is highly consistent with the cooling scheme proposed in this paper, where coolant flows through the internal interlayer spaces of the sphere.

Third, the circulation operational logic of continuous refueling. The HTGR employs an on-load refueling method, where fuel pebbles are loaded from the top of the core and discharged from the bottom by gravity, undergoing approximately 15 passes through the core before reaching the design burn-up depth [8]. This "circulation, renewal, continuous operation" model is structurally analogous to the brain-inspired operational logic proposed in this paper, where data flows continuously within the sphere and is processed locally.

Fourth, the mathematical framework of multiscale physics. A pebble-bed core is a typical multiscale physical system: the macroscopic scale involves the random packing of hundreds of thousands of pebbles, the mesoscopic scale encompasses tens of thousands of TRISO (tristructural isotropic) fuel particles within each pebble, and the microscopic scale concerns the physical processes of nuclear reactions. The academic community has developed mature mathematical modeling frameworks for this system, such as the open-source MOOSE (Multiphysics Object-Oriented Simulation Environment) framework developed at Idaho National Laboratory, which is also applicable to the thermohydraulic simulation of computing pebble cores.

1.4 From "Microscopic Integration" to "Macroscopic Restructuring": The Divergence of Two Paths

In the face of physical limits, the mainstream semiconductor industry's response strategy is "microscopic three-dimensional integration": by leveraging technologies such as through-silicon vias (TSVs) and hybrid bonding, multiple chip layers are vertically stacked to form micron-scale three-dimensional structures, while keeping the chip's planar footprint unchanged. Representative examples of this approach include TSMC's 3D-SoC, IMEC's CMOS 2.0 [12], and DARPA's 3D heterogeneous integration program [15].

However, this approach does not fundamentally solve the three physical red lines. While vertical stacking increases transistor density, it also exacerbates the thermal density problem — the heat conduction path from the intermediate layers to the heat dissipation surface becomes longer, with greater thermal resistance. For a 3D chip dissipating 100 W, the heat at its center must traverse layer upon layer of silicon and transistors to reach the heat dissipation surface. This "sleeping under a thick quilt" thermal structure means that the interior of the stacked chip will inevitably overheat.

The "macroscopic restructuring" path proposed in this paper adopts a fundamentally different engineering decision: proactively abandoning the relentless pursuit of "thin, light, short, small," and instead using a fist-sized or even larger macroscopic spherical volume to obtain ample three-dimensional routing space, embedded vascular cooling networks, and the

physical space required for integrating optical interconnect devices. This is conceptually analogous to ENIAC, the first electronic computer, which used an entire room to accommodate 18,000 vacuum tubes — sacrificing volume to solve at that time the heat dissipation problem. The difference is that this time, it is a deliberate and optimized spatial design rather than a forced compromise.

2 Architecture Overview: Design Philosophy and Fundamental Principles of the Computing Pebble Core

2.1 Definition and Core Concept of the "Computing Pebble Core"

This paper formally proposes the concept of the "Computing Pebble Core" (CPC). Its core idea can be summarized by four "native integrations":

Native Integration of Computing and Memory: Breaking the physical separation of the Von Neumann architecture, within the three-dimensional spherical space, each computing node carries local SRAM or eDRAM storage (capacity approximately 1–10 MB), forming an independent autonomous computing unit with integrated compute and memory. Data no longer requires long-distance travel; each computation is completed within, or within a few hops of, the node where the data resides.

Native Integration of Electronic Computation and Optical Communication: Long-distance inter-node communication within the sphere is carried by an optical interconnect network, while the micro-scale logic operations within each computing node are still completed by electronic circuits. This is an asymmetric hybrid-signal architecture that fully exploits the advantages of photons in long-distance transmission (low latency, low power) and the advantages of electrons in micro/nano-scale logic switching (high density, high energy efficiency).

Native Integration of Heat Dissipation and Structural Framework: Drawing inspiration from the human vascular system, the hollow pillars between layers within the sphere serve both as mechanical structural support and as hollow liquid-cooling channels. The coolant undergoes forced convective circulation within these "skeleton-blood vessels,"

efficiently carrying heat away from every corner of the three-dimensional space. The cooling structure is not an additional "heat sink" but an inseparable part of the sphere itself.

Isomorphism of Spatial Topology and Algorithmic Models: The computing nodes within the sphere are connected via a three-dimensional network topology combining latitude-longitude lines and radiallayered, whose physical topology naturally suits graph-based neural network algorithms, high-dimensional matrix operations, and sparse spiking communication patterns. The physical architecture is no longer merely a passive carrier for algorithms but forms a "structure-function" isomorphic mapping with the algorithm, achieving true hardware-software co-optimization.

2.2 Architecture Topology Selection: The Pareto Optimality of the Regular Icosahedron

The topology selection for a three-dimensional computing architecture is the primary determinant of its engineering feasibility. This section systematically compares six candidate topologies — sphere, cube, cylinder, regular octahedron, regular dodecahedron, and regular icosahedron — from both mathematical and engineering perspectives, using three core metrics: manufacturing compatibility, spherical approximation ratio, and mechanical stability.

Definition of Comparison Dimensions:

- **Spherical Approximation Ratio:** Measured as the ratio of the topology's surface area to that of an equal-volume sphere. A perfect sphere scores 1.00. This metric determines the equidistance of internal node spatialarrangement and the uniformity of mechanical load bearing.

- **Facet Planar Manufacturability:** Whether the constituent facets of the topology are all congruent planar polygons that can be directly fabricated on standard wafers. This is a key factor determining mass production efficiency.

- **Facet Congruence:** Whether all facets have identical shape and dimensions. Congruent facets mean only one set of photomasks is needed to manufacture all facets, reducing mask costs by over an order of magnitude.

- Mechanical Compression Uniformity: The uniformity of external pressure distribution across facets, measured by the ratio of maximum to minimum stress. The closer this ratio is to 1, the higher the reliability of the structure in extreme mechanical environments such as deep space and deep sea.

- Inter-Facet Assembly Complexity: The number of facets and inter-facet bonding surfaces, which determines the complexity of the assembly process and cumulative alignment error.

- Wafer Area Utilization: The area utilization efficiency when arranging a single facet on a circular wafer.

Topology Comparative Analysis:

Table 1. Topology comparison of candidate 3D architectures.

Topology	Facets	Spherical Approx.	Facet Congruence	Planarity	Stress Ratio	Wafer Utilization	Overall
Cube	6	0.81	Congruent squares	Planar	2.8	~85%	Simplest mfg; worst sphericity; uneven mechanics
Regular Octahedron	8	0.85	Congruent triangles	Planar	2.1	~70%	Few facets; insufficient sphericity
Regular Dodecahedron	12	0.91	Congruent pentagons	Planar	1.5	~58%	Good sphericity; low wafer utilization
Regular Icosahedron	20	0.94	Congruent triangles	Planar	1.2	~65%	Excellent sphericity; optimal mechanics; good utilization
Cylinder	3	0.83	Non-congruent	Partially curved	3.0+	N/A	Curved surface; infeasible for mfg
Perfect Sphere	∞	1.00	—	Curved	1.0	N/A	Best sphericity;

							not mass- producible
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Proof of Pareto Optimality for the Regular Icosahedron:

From the table above, it can be seen that the regular icosahedron constitutes a Pareto-optimal solution across the four dimensions of spherical approximation ratio, facet congruence, planarity, and mechanical uniformity: no other topology outperforms the regular icosahedron in any single dimension without being significantly worse in others. Specifically:

Spherical Approximation Ratio of 0.94: The highest among all Platonic solids, meaning the spatial arrangement of nodes inside the sphere closely approaches an ideal spherical surface, with the advantage of equidistance almost fully preserved. The average path length between nodes scales with $N^{(1/3)}$ (where N is the total number of nodes), rather than $N^{(1/2)}$ for planar grids. At a scale of 1,000 nodes, the average path length is reduced by approximately 3 times.

20 Congruent Equilateral Triangular Facets: All facets are identical in shape, requiring only a single set of photomasks to manufacture all 20 facets. Compared to non-congruent facet designs (e.g., spherical latitude-longitude division requiring dozens of different mask sets), mask costs are reduced by approximately 95%. This is the fundamental guarantee of mass production economics.

Standard Wafer Compatibility and Area Utilization: The maximum side length of each equilateral triangular facet can be designed to within 0.5 times the wafer diameter, ensuring that a single 12-inch wafer (usable diameter approximately 290 mm) can accommodate at least 4 to 6 complete facets at once. The area utilization of triangular arrangement on a circular wafer is approximately 65%, which, although lower than the 85%–90% of rectangular chips, is a fully acceptable cost given the dimensional benefits of 3D integration — the computing density of a regular icosahedral sphere is hundreds to thousands of times that of planar chips at the same technology node, more than compensating for the gap in wafer utilization.

Optimal Mechanical Stability: When the regular icosahedron is subjected to uniform external pressure, the stress distribution is the most uniform, with a maximum-to-minimum stress ratio of only 1.2, far lower than the cube's 2.8 and the regular octahedron's 2.1. This property makes it naturally suitable for extreme mechanical environments such as deep-sea high pressure (extremely low stress concentration factor) and rocket launch vibration (high and uniform structural natural frequency).

Inter-Facet Angular Error Control: The dihedral angle between adjacent faces of the regular icosahedron is 138.19° . Through precision mechanical alignment and hybrid bonding processes, the inter-facet angular error can be controlled to within 0.1° , satisfying the sub-micron alignment precision requirements for optical waveguide interfacing.

Based on the above analysis, the regular icosahedron is established as the standardized topology for the HCC architecture. The far-reaching significance of this choice is that it provides a standardized topological substrate for all three-dimensional macroscale computing architectures, just as the rectangular grid does for planar chips. All subsequent discussions on manufacturing processes, interconnect networks, and cooling systems are based on this topology.

2.3 Demon-Work Ball-Style Multi-Layer Nested Hollow Structure

The physical form of the Computing Pebble Core is a demon-work ball-style multi-layer nested hollow sphere based on the regular icosahedral framework.

Specifically, the sphere consists of multiple layers of regular icosahedral frameworks, supported between layers by beams and columns, forming hollow interlayers of approximately hundreds of microns to several millimeters. The beams and columns themselves are hollow structures, housing coolant or routing inside. The innermost core contains a dense cluster of computing nodes, the middle layers comprise storage and routing layers, and the outermost layer contains optical transceiver and external interface layers. Vertical signal transmission between layers is achieved through electrical connections and optical paths within the through beams and columns.

This "demon-work ball" hollow structure yields three physical benefits:

Geometric Multiplication of Heat Dissipation Area: Like the alveoli in human lungs, the total surface area of the internal channel walls can reach hundreds or thousands of times the projected area of the sphere, providing an enormous exchange area for convective heat transfer. Assuming a sphere diameter of 10 cm with 5 nested layers, each layer having 100 hollow beams/columns of 1 mm diameter, the total internal channel wall surface area is approximately 800 times the projected area of the sphere.

Average Equidistance of Shortest Paths: In the icosahedral topology, the distance from any point to the sphere center is approximately equal, and the average path length between nodes is far shorter than in a planar layout with the same number of nodes. For a planar grid with N nodes, the average path length scales with $\sqrt{N}$; in the icosahedral 3D network, it scales with $\sqrt[3]{N}$ — this is an inherent mathematical advantage of spherical geometry. When $N = 1,000$, the average path length of the 3D spherical topology is only approximately 1/3 that of the planar topology.

Spatial Abundance for 3D Routing: The hollow interlayers and beam/column surfaces provide ample three-dimensional space for the independent layout of optical paths, electrical circuits, and cooling channels, completely avoiding the problem of "signal lines, power lines, and ground lines crowded together" in planar chips. The three types of physical channels each occupy their own space, without mutual interference.

2.4 Full Mapping of Nuclear Core Design Principles: From Four Dimensions to Four Subsystems

At this point, we can establish a complete mapping of design principles between the pebble-bed HTGR and the Computing Pebble Core. The HTGR core is formed by the random close packing of hundreds of thousands of fuel pebbles, with helium coolant undergoing forced convective circulation through the inter-pebble pore spaces. The core achieves continuous loading and discharge of fuel pebbles through gravity, and each TRISO fuel particle is an independent micro-reaction unit.

The four principles mapped to the computing system constitute the most fundamental design axioms of the HCC architecture:

Axiom I (Energy Conservation Mapping): The heat from the nuclear core is carried away by convective flow of helium through-fluid; the computation heat from the CPC is carried away by forced convective circulation of interlayer coolant. The heat dissipation path evolves from "external attachment" to "internal integration," from "passive" to "active."

Axiom II (Material Circulation Mapping): Fuel pebbles in the nuclear core are continuously renewed through gravity circulation; data packets in the CPC are continuously circulated through the optical network and processed locally, achieving "data does not relocate; computation stays at home."

Axiom III (Unit Autonomy Mapping): Each TRISO fuel particle in the nuclear core is an independent micro-fission unit; each compute-in-memory node in the CPC is an independent autonomous computing unit with local storage, local logic, and an independent communication interface.

Axiom IV (Spatial Topology Mapping): The nuclear core adopts the optimal spherical close-packing configuration; the CPC adopts the optimal icosahedral 3D network topology, achieving Pareto optimality between manufacturing feasibility and spherical approximation.

2.5 Three-Dimensional Spherical Coordinate Addressing System

Within the sphere formed by the regular icosahedron, each computing node is assigned a unique three-dimensional spherical coordinate address. The coordinate system design draws inspiration from geographic coordinate systems and is specifically adapted to the geometric characteristics of the regular icosahedron:

Radial coordinate ρ : Indicates the layer number of the node, from the innermost layer ($\rho = 0$) to the outermost layer ($\rho = R$), corresponding to the nesting levels of the sphere. The

radial spacing between layers is jointly determined by the microchannel interlayer thickness and signal transmission distance constraints, with typical values of 2 to 5 mm.

Polar angle coordinate θ : Indicates the latitudinal position of the node on its layer. The 20 triangular faces of the regular icosahedron are further subdivided into a latitude-longitude grid, and the subgrid where each node is located is uniquely determined by its polar angle θ . Under regular subdivision, the number of discrete values for θ corresponds to the latitudinal division precision of the triangular faces on that layer.

Azimuthal coordinate φ : Indicates the longitudinal position of the node on its layer. Together with the θ coordinate, it jointly determines the two-dimensional position of the node on a given layer. The number of discrete values for φ corresponds to the longitudinal division precision of the triangular faces on that layer.

The network address of each computing node is uniquely determined by its triplet (ρ , θ , φ). This addressing scheme offers the following advantages: the address space corresponds strictly to the physical space, so route computation requires no lookup tables; node addresses directly encode their spatial positions, supporting coordinate-based deterministic routing; the radial coordinate naturally supports hierarchical routing strategies, decomposing the global routing problem into two sub-problems: intra-layer routing and inter-layer routing.

3 Material System: The Semiconductor Foundation for Three-Dimensional Computing

3.1 The Limits of Silicon and the Rise of Wide Bandgap Semiconductors

Silicon has dominated the semiconductor industry for more than half a century based on three irreplaceable physical advantages: single-crystal perfection, a natural surface insulating layer (SiO_2), and extremely low cost. However, in the context of three-dimensional macroscale integration, silicon's advantages diminish while its disadvantages become prominent — silicon has a bandgap of only 1.12 eV, its intrinsic carrier concentration rises sharply with temperature, and device performance deteriorates rapidly or fails above 150°C.

Wide bandgap semiconductors such as silicon carbide (SiC) and gallium nitride (GaN) have bandgaps of 3.3 eV and 3.4 eV, respectively — approximately three times that of silicon. The table below compares the key physical parameters of the three materials:

Table 2. Key physical parameters of semiconductor materials (Si, 4H-SiC, GaN).

Parameter	Si	4H-SiC	GaN
Bandgap (eV)	1.12	3.26	3.4
Critical Breakdown Field (MV/cm)	0.3	2.8	3.5
Thermal Conductivity (W/m·K)	150	490	250
Electron Saturation Velocity (10^7 cm/s)	1.0	2.0	2.5
Max Operating Temperature (°C)	150	500	500
Radiation Resistance (relative)	1×	100×	100×

As evident from the table, integrated circuits based on GaN/SiC can operate stably at temperatures of 300–500°C, fundamentally solving the most intractable "core burnout" problem in three-dimensional computing structures. At the same time, their two-orders-of-magnitude advantage in radiation resistance makes this architecture naturally suitable for extreme application scenarios such as deep space exploration and nuclear environments. In deep space exploration, the radiation dose rate in Jupiter's orbit can be more than 1,000 times that of geosynchronous orbit; in post-accident nuclear power plant monitoring, the ambient radiation dose rate can surge by 5 to 6 orders of magnitude in a short time. These are scenarios where traditional silicon-based chips are completely incapable of functioning.

3.2 Industrial Breakthrough in GaN-on-SiC Monolithic Heterogeneous Integration

The heterogeneous integration of GaN and SiC has long been plagued by two core challenges: first, the high cost and small size limitations of large-diameter SiC substrates;

second, the lattice mismatch (approximately 3.5%) and thermal mismatch in achieving high-quality GaN epitaxial growth on SiC substrates.

In August 2025, Shenzhen Pinghu Laboratory achieved the world's first commercial 8-inch 4° off-angle 4H-SiC substrate with high-quality AlGaIn/GaN heterostructure epitaxy [13]. Through optimization of the 4° off-angle design, this technology reduced threading dislocation density to below 10^8 cm^{-2} and increased two-dimensional electron gas (2DEG) mobility to above $2,000 \text{ cm}^2/\text{V}\cdot\text{s}$, solving the material defect and thermal dissipation bottlenecks at the crystal growth level. This breakthrough marks the transition of GaN-on-SiC heterogeneous integration from the laboratory to the engineered mass production stage on 8-inch wafers.

Meanwhile, HRL Laboratories in the United States announced in 2026 a manufacturing milestone on a 40 nm GaN-on-SiC process, with plans to advance high-volume manufacturing through MACOM while maintaining small-volume engineering runs and multi-project wafer (MPW) services for qualified customers [14]. DARPA and the Texas state government have invested \$1.4 billion in building a 3D heterogeneous integration experimental fab, aiming to achieve layer-by-layer stacking and interconnection of multiple semiconductor materials — including silicon, GaN, and SiC — within a single fab [15]. According to DARPA projections, silicon-based chip stacking can improve performance by up to 30 times compared to traditional 2D designs, while heterogeneous semiconductor materials are expected to achieve a 100-fold improvement.

These landmark industrialization events indicate that GaN-on-SiC monolithic heterogeneous integration has emerged from the laboratory and is on the cusp of engineered mass production. The HCC architecture has obtained strong industrial support at the material level.

3.3 Diamond Heat Dissipation Shell: The "Golden Jade Garment" Design

On the outermost layer of this architecture, a synthetic diamond heat dissipation shell is grown using chemical vapor deposition (CVD). Diamond has the highest thermal conductivity of any known material — up to $2,000 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, which is 4–5 times that of

copper, approximately 4 times that of SiC, and approximately 13 times that of silicon. In addition, diamond has an extremely wide bandgap (approximately 5.5 eV), very high current-carrying capacity, and excellent radiation resistance [16].

To overcome the challenge that traditional CVD diamond thin film deposition temperatures (typically above 800°C) would damage already-fabricated metal interconnects and device structures, this design draws on the latest research results from TSMC and National Cheng Kung University: depositing high-quality diamond thin films at BEOL-compatible temperatures of 450°C or below. Experimental data show that diamond thin films prepared with this technology have a stable thermal conductivity above $300 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, and as grain size further increases to above 5 μm , thermal conductivity can continue to rise to above $1,000 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ [17].

This diamond shell works in synergy with the liquid cooling network in the internal hollow interlayers to form a complete three-level thermal management system of "heat spreading — heat transport — heat rejection":

Level 1 (Heat Spreading): The diamond shell is responsible for rapidly "spreading" the large amount of heat generated by internal hotspots across the entire sphere surface, spatially homogenized localized hotspots. Diamond's extremely high thermal conductivity ensures this process is completed on a microsecond timescale.

Level 2 (Heat Transport): The coolant in the hollow interlayers, through forced convective circulation, efficiently transports heat from the internal layers to the inner surface of the diamond shell.

Level 3 (Heat Rejection): An external heat exchanger and compressor system, forming a "mechanical lung," ultimately rejects the heat carried by the coolant to the ambient atmosphere or cooling water.

4 Communication System: Three-Tier Optoelectronic Hybrid Interconnect Architecture

4.1 Current Industrial Status of Optical Interconnects

Optoelectronic hybrid computing is no longer a laboratory concept. From 2025 to 2026, silicon photonic integration achieved multiple milestone industrial breakthroughs.

TSMC's COUPE (Compact Universal Photonic Engine) has entered 65 nm mass production, adopting a three-tier architecture that stacks approximately 220 million transistors and 1,000 integrated optical elements. Compared to traditional copper interconnects, COUPE uses 3D-stacked photonic chips to reduce transmission distance to the micron scale, improving energy efficiency by 4 times, reducing latency by 10 times, and with deep integration, energy efficiency can be improved by 10 times. Single-fiber data rates support 1,024 Gbps, and integration into CoWoS packaging as co-packaged optics (CPO) is planned for 2026 [18].

California-based startup Lightmatter's Passage M1000 platform goes even further, achieving 114 Tbps total bidirectional optical bandwidth, with single-fiber bidirectional bandwidth of 800 Gbps (16-wavelength bidirectional link) and energy efficiency of 4.6 pJ/bit. Its L200 3D CPO series offers 32 Tbps and 64 Tbps versions, providing a 5–10× improvement over existing solutions, with single-chip total I/O bandwidth exceeding 200 Tbps, capable of reducing advanced AI model training time by up to 8 times [19][20][21].

4.2 Three-Tier Optoelectronic Division of Labor: Transforming the "Conversion Tax" into an Architectural Advantage

The overhead of optoelectronic conversion — the energy consumption and time delay incurred when electrical signals are converted to optical signals and then back to electrical signals — is a cost that any optoelectronic hybrid system must confront. The most advanced current silicon photonic transceivers achieve energy efficiency of approximately 1–2 pJ/bit (for optical transmission only), with complete electrical-optical-electrical conversion energy efficiency of approximately 5–10 pJ/bit and a conversion delay of approximately 1–2 ns.

This paper proposes a three-tier optoelectronic division-of-labor system that uses different signal carriers at different physical scales based on communication distance and bandwidth requirements, minimizing the number of optoelectronic conversions:

Tier 1: Intra-Layer Short-Distance Communication (distance < 1 mm) — Pure Electrical Interconnect.

Communication between adjacent computing nodes within the same icosahedral facet typically ranges from hundreds of microns to 1 mm. At this scale, copper interconnect energy efficiency (< 1 pJ/bit) and latency (< 0.1 ns) are both superior to optoelectronic conversion schemes. Therefore, intra-layer communication exclusively uses traditional copper interconnects, completely bypassing the optoelectronic conversion stage. This decision ensures that the vast majority of local computing communication (estimated at 60%–70% of total communication) avoids optoelectronic conversion overhead.

Tier 2: Inter-Layer Medium-Distance Communication (distance 1–10 mm) — Hybrid Interconnect.

Radial vertical communication between different layers typically ranges from 1 to 10 mm. At this scale, a hybrid interconnect scheme is adopted, with TSV electrical connections and optical waveguides operating in parallel. High-frequency, high-bandwidth data streams (such as intermediate result synchronization for matrix operations, gradient updates for model parameters) take the optical path; low-frequency, low-bandwidth control signals (such as routing table updates, clock synchronization, fault detection signals) take the electrical path. Optoelectronic conversion at this tier is "selective," performed only for data packets that genuinely require high-speed transmission. A typical data stream allocation strategy is: data packets with bandwidth requirements exceeding 1 Gbps take the optical path; the rest take the electrical path.

Tier 3: Cross-Layer Long-Distance Communication (distance > 10 mm) — Pure Optical Interconnect.

For long-distance communication spanning most of the sphere (e.g., from a computing node on one side of the sphere to a node on the opposite side), the distance exceeds 10 mm. At this scale, the RC delay of electrical interconnects has already exceeded 100 ps and grows exponentially with distance, while transmission power consumption rises sharply. In contrast, the transmission energy consumption of optical interconnects remains

almost unchanged with distance (depending only on waveguide loss, typically 1–3 dB/cm). Therefore, all cross-region, cross-layer long-distance communication is mandatory routed through the optical path.

Introduction of All-Optical Switching Nodes:

To further reduce the number of optoelectronic conversions, eight all-optical switching nodes are deployed at keyhub positions within the sphere: one at the sphere center (corresponding to $\rho = 0$, the core layer), four on the equatorial plane (evenly distributed on the $\theta = \pi/2$ equatorial plane), and four distributed above and below the orthogonal polar axis (two each). These eight all-optical switching nodes can directly perform routing and switching of data packets within the optical domain, without the need for conversion to electrical signals and back to optical signals. Once a data packet enters an all-optical switching node, the microring resonator array performs wavelength-division multiplexing route selection based on the target optical wavelength of the packet, remaining in the optical domain throughout.

Each all-optical switching node supports 128 input ports and 128 output ports, with a single-port data rate of 1.6 Tbps, for a total switching capacity of 204.8 Tbps. For the 2040 production model with 10,000 nodes, the total capacity of 8 all-optical switching nodes reaches 1.6 Pbps, sufficient to support peak communication traffic. Future expansion of system scale is possible by increasing the number of all-optical switching nodes and port density.

With the introduction of all-optical switching nodes, the average number of optoelectronic conversions for a typical data packet is reduced from 5–10 times to 2–3 times, reducing the cumulative overhead of optoelectronic conversion by approximately 60%–70%.

Staged Optoelectronic Conversion Energy Efficiency Targets:

Table 3. Staged optoelectronic conversion energy efficiency targets.

Time Node	O/E Conversion Efficiency (pJ/bit)	Technology Path
2027 Prototype	≤ 10	Based on existing Si photonic transceivers
2030 Prototype	≤ 5	Optimized MRR design;

		subwavelength grating couplers
2035 Engineering Prototype	≤ 2	Plasma-enhanced photodetectors; reduced drive voltage
2040 Production Model	≤ 1	Novel electro-optic materials (e.g., TFLN); sub-pJ/bit

4.3 Spherical Routing Algorithm: A Design Unique to Icosahedral Topology

The regular icosahedral spherical topology possesses unique mathematical properties for routing algorithms that differ from cubic topologies. This paper proposes a three-tier hierarchical routing architecture, with computational complexity ranging from $O(1)$ to $O(\log N)$, enabling a smooth upgrade from deterministic shortest-path to congestion-aware adaptive routing.

Bottom-Layer Routing: Coordinate-Based Deterministic Routing — Computational Complexity $O(1)$.

The most basic routing layer adopts an address resolution method similar to GPS satellite positioning. Each data packet carries the spherical coordinates of the target node (ρ_{target} , θ_{target} , ϕ_{target}). The current node directly computes the next-hop direction based on the difference between its own coordinates and the target coordinates. For intra-layer routing (same ρ value), next-hop selection follows the "great circle shortest arc" principle — on the icosahedral triangular grid, select the adjacent node that minimizes the spherical arc length. This computation has a constant complexity of $O(1)$, making it suitable for direct hardware implementation.

Middle-Layer Routing: Great Circle Shortest-Path Routing — Computational Complexity $O(\log N)$.

For cross-layer long-distance communication, the middle-layer routing computes the great circle path between the source and target nodes — the shortest path between two points on a sphere. Drawing on civil aviation route planning and inter-satellite link routing algorithms in satellite communications, the great circle path is discretized into a number of

hops on the icosahedral grid. Since the 20 triangular faces of the regular icosahedron can be recursively refined into latitude-longitude grids of arbitrary precision, great circle computation can be completed in logarithmic-linear time.

Top-Layer Routing: Congestion-Aware Adaptive Routing — Drawing on Starlink Satellite Network Protocols.

When network load is high, some links may experience congestion. The top-layer routing introduces an adaptive routing protocol inspired by SpaceX's Starlink low-earth-orbit satellite network. Starlink has already demonstrated the feasibility of large-scale three-dimensional spatial dynamic routing in a network of tens of thousands of satellites — each satellite dynamically adjusts data packet forwarding paths based on real-time link status information, bypassing congested or failed nodes [22]. This protocol can be directly ported to this architecture, simply replacing the three-dimensional orbital coordinates of satellites with the spherical coordinates of computing nodes.

Unique Routing Advantages of Spherical Topology:

Compared to cubic three-dimensional networks, the regular icosahedral spherical topology offers two unique routing advantages:

Boundary-Free Load Balancing: Cubic 3D networks have "edge nodes," "edge nodes," and "corner nodes," which have fewer neighbors than internal nodes, leading to uneven load distribution — congestion rates at edge nodes can be more than 3 times those of central nodes. In the regular icosahedral spherical topology, all nodes have the same number of neighbors (under regular grid subdivision), resulting in naturally uniform load distribution.

Fault Tolerance of Infinite Equivalent Paths: In the regular icosahedral spherical topology, there exist numerous nearly equivalent paths between any two nodes (along different great circle directions). When a link or node fails, data packets can automatically reroute without requiring global recalculation of routing tables. Quantitatively, in a regularly subdivided latitude-longitude grid of the regular icosahedron (precision of 10 nodes per edge, total of approximately 1,000 nodes), there exist on average approximately 8–15 nearly equal-length paths between any pair of nodes. This means that even if 3–5 intermediate nodes fail simultaneously, data packets can still find an alternative shortest path with almost unchanged

latency. In a cubic grid of equivalent scale, the average number of equal-length shortest paths is only 2–4, representing significantly insufficient fault tolerance.

5 Cooling System: Vascular Embedded Microfluidic Circulation

5.1 Latest Breakthroughs in "Electronic Blood" Industrial Implementation

The vision of "coolant flowing like blood through the blood vessels of a chip" has transitioned from science fiction to engineering reality in 2025–2026.

Microsoft, in collaboration with Swiss startup Corintis, jointly announced the etching of micron-scale fluid channels directly inside chips, allowing coolant to flow into the most core positions within the chip body to carry away heat. Test data showed that this solution improved cooling efficiency by 3 times compared to the most advanced solutions at the time, and reduced the maximum internal temperature rise of the GPU by 65% [23]. This represents the most direct engineering validation of the "embedded microchannel" concept.

In 2025, Professor Song Bai's team at Peking University published a paper in *Nature Electronics*, innovatively proposing a "manifold-microjet-serpentine microchannel" composite embedded microfluidic structure, using single-phase water as the coolant, achieving an ultra-high heat flux density of 3,000 W/cm² for chip cooling — setting a world record. This technology is fully compatible with integrated circuit manufacturing processes, does not rely on special materials, and combines high cooling performance with low pumping power consumption — at a heat flux density of 3,000 W/cm², the chip temperature rise is only approximately 60°C, and pumping power consumption accounts for less than 5% of cooling power [24].

Meanwhile, Professor Wang Yucheng's team at Northwestern Polytechnical University proposed a data-driven optimization framework based on an enhanced genetic algorithm, demonstrating that algorithm-optimized liquid metal microfluidic cooling systems can effectively dissipate heat for chips with power consumption up to 800 W while maintaining stable thermal performance [25].

5.2 Fractal Vascular Network Topology Design within the Sphere

This architecture deeply integrates the cooling channels into the structural framework of the sphere. Drawing on the fractal vascular network theory of mammalian circulatory systems — a hierarchical branching structure from the aorta to capillaries, achieving maximum convective heat transfer area with minimal pumping work — the coolant channels within the sphere exhibit the following topology:

Aorta-Level: After entering the sphere from the external pump interface, the coolant first enters 6 main pipes (corresponding to the 6 fivefold symmetry axes of the regular icosahedron), with diameters of approximately 1–2 mm and flow velocities of approximately 1–2 m/s.

Artery-Level: After each main pipe enters each regular icosahedral facet, it branches into 3–5 secondary pipes, with diameters of approximately 0.5–1 mm, leading to each layered interlayer.

Capillary-Level: Within each interlayer, the secondary pipes further branch into dozens of micro-pipes, with diameters of approximately 50–200 μm , embedded inside the hollow beams and columns, in close proximity to the computing nodes.

Vein-Level: After absorbing heat, the micro-pipes merge into secondary return pipes, ultimately returning to the external heat exchanger, completing one cycle.

The theoretically optimal branching level of the fractal vascular network can be determined by Murray's Law: at each branching point, the cube of the parent vessel radius equals the sum of the cubes of the child vessel radii, i.e., $r_{\text{parent}}^3 = \sum r_{\text{child}}^3$ [26]. A fractal network following Murray's Law achieves maximum heat transfer area and heat transfer capacity with minimal fluid resistance and pumping power consumption.

Coolant Type Selection:

Table 4. Coolant type selection and applicable scenarios.

Coolant Type	Thermal Cond. (W/m·K)	Specific Heat (J/kg·K)	Applicable Heat Flux (W/cm ²)	Application Scenario
Deionized Water	0.6	4180	≤1000	Prototype validation
Ga-In-Sn Liquid Metal	30	400	1000–3000	High-power engineering prototype
Dielectric Coolant (e.g., Novec)	0.07	1100	≤500	Electrical insulation required

5.3 Thermal-Fluidic-Electrical-Optical Four-Field Coupled Simulation Platform

The nuclear engineering field has developed mature numerical analysis frameworks for multiphysics coupling of neutron physics and thermal-hydraulics. The multiscale structure of the pebble-bed core — the macroscale random close-packed pebbles, the mesoscale inter-pebble pore flow channels, and the microscale coated fuel particles — is highly isomorphic with the physical scale stratification of this architecture. However, there are essential differences between computing systems and nuclear reactors that prevent simple transplantation:

The heat source in a nuclear reactor (fission reaction) is approximately uniformly distributed in space, whereas the heat source in a computing chip is highly dynamic and non-uniform — the temperature difference between hotspots and cold spots can exceed 50°C, and hotspot locations move in real time with the workload.

The coolant flow in a nuclear reactor is steady-state under normal operating conditions, whereas the cooling demand in a computing system is transient — an AI inference task may require 500 W of cooling capacity one second and drop to 50 W the next.

Nuclear engineering tools do not consider electromagnetic interference (EMI) and signal integrity issues, which are primary design constraints in high-frequency optoelectronic hybrid systems.

Brief EMI Analysis: Due to the hierarchical routing strategy adopted by this architecture — electrical signals travel through the interior of beams and columns, optical signals through the interlayer spaces, and the operating voltages of wide bandgap semiconductors (GaN HEMT typical operating voltage 5–10 V, far lower than the 0.7–1.2 V threshold-sensitive region of silicon-based high-performance devices) and current densities are both lower than those of equivalent-performance silicon-based devices — EMI levels are more than an order of magnitude lower than those of traditional planar chips. Electromagnetic simulation primarily focuses on crosstalk of high-frequency electrical signals passing through TSVs, as well as resonances in the power distribution network. Since the power and ground layers within the sphere can adopt a complete spherical metal shielding layer (utilizing the outer metalization of the regular icosahedron), electromagnetic radiation leakage can be effectively suppressed.

In response to the above differences and requirements, this paper proposes a three-phase simulation platform development roadmap: [See text for Phase 1–3]

Phase 1 (before 2027): Based on the thermal-hydraulic solver of Xi'an Jiaotong University's NECP-Panda code, add dynamic heat source terms and transient flow models. Replace the neutron diffusion equation in the solver with a heat diffusion equation, and replace the uniform heat source with a non-uniform heat source function adjustable in real time based on computational load, providing preliminary adaptation to the dynamic, non-uniform hotspot characteristics of computing chips.

Phase 2 (before 2030): Integrate ANSYS Icepak's electronic cooling simulation module and Ansys Lumerical's photonics simulation module to achieve weak coupling of three physical fields: thermal-fluid, electromagnetic, and optical. Weak coupling means the solvers for each physical field operate independently, exchanging boundary condition data through interfaces, suitable for rapid iteration during the prototype phase.

Phase 3 (before 2035): Develop a dedicated thermal-fluid-electrical-optical four-field fully coupled solver based on an open-source multiphysics framework (such as MOOSE [27]). Leverage the regular grid characteristics of the icosahedral spherical structure for geometric modeling simplification and parallel computing optimization, with the target of improving

simulation speed by more than 100 times compared to general-purpose commercial software, reducing a single fully coupled simulation from days to minutes.

Collaboration Framework: It is recommended to establish a joint simulation laboratory with the Nuclear Engineering Computational Physics Laboratory (NECP-Panda) at Xi'an Jiaotong University, the State Key Laboratory of Turbulence and Complex Systems at Peking University (CFD), and the Institute of Optics and Electronics at the Chinese Academy of Sciences (optical simulation), sharing algorithms, data, and validation cases.

6 Computing Paradigm: Three-Dimensional Implementation of Brain-Inspired Compute-in-Memory

6.1 Breaking the Von Neumann Bottleneck with 3D Compute-in-Memory

Processing-in-Memory (PIM) is one of the most promising computing paradigms of the post-Moore era. Its core idea is to physically integrate computing units and storage units in close proximity, fundamentally eliminating the overhead of shuttling data back and forth between memory and processors.

Currently, 3D-stacked PIM technology has entered the engineering promotion stage. Micro-Nano Core's 3D-CIM technology, combining SRAM compute-in-memory with DRAM 3D stacking, completes computation within the memory array. In September 2025, under the guidance of the Ministry of Industry and Information Technology, Micro-Nano Core, as the lead unit of the RISC-V Compute-in-Memory Application Group, initiated the development of the world's first RISC-V CIM standard in Xiaoshan, Hangzhou, uniting over 20 leading industry chain enterprises to promote the construction of an autonomous AI chip ecosystem [28]. TsingMicro Intelligence, by three-dimensionally stacking computing units and storage units — analogous to directly bonding processors and memory — has constructed a 3D DRAM CIM architecture with ultra-high bandwidth [29].

However, these solutions still operate at the microscopic scale through vertical stacking, utilizing only one direction of three-dimensional space (the vertical direction), without fully exploiting all three degrees of freedom. This architecture pushes this concept to

its extreme: within the three-dimensional space of the regular icosahedral sphere, computing units and storage units are arranged in an interleaved fashion across each facet — each computing node is equipped with local SRAM or eDRAM storage (capacity approximately 1–10 MB), while frequently interacting node pairs are physically adjacent in space. This principle of "spatial proximity" realizes, in the silicon world, the arrangement of neurons in the human brain — functionally related neurons tend to be physically adjacent in the cortex to reduce axon length and conduction delay.

6.2 Three-Dimensional Hardware Implementation of Brain-Inspired Spiking Neural Networks

Spiking Neural Networks (SNNs) process information through event-driven sparse spikes, consuming power only when spikes occur, achieving far higher energy efficiency than traditional synchronous clock-driven deep neural networks. This sparsity naturally suits large-scale three-dimensional networks — when the majority of nodes are in a silent state most of the time, the total power consumption and thermal pressure of the sphere are significantly reduced. According to estimates, in typical image recognition tasks, the activity rate of SNNs (i.e., the proportion of neurons in an activated state at any given time) is typically below 5% [30], meaning 95% of nodes are in a low-power silent state 95% of the time. The event-driven nature of SNNs enables energy efficiency 10–20 times higher than traditional synchronous clock networks. This advantage has been incorporated into the quantitative performance model in Section 9 of this paper and is one of the core supports for the 2040 production model achieving an energy efficiency ratio of 9 TFLOPS/W.

In August 2025, the State Key Laboratory of Brain-Machine Intelligence at Zhejiang University released the next-generation neuromorphic brain-like computer "Wu Kong" (Darwin Monkey), equipped with 960 self-developed Darwin-3 brain-inspired computing chips, supporting over 2 billion spiking neurons and 100 billion neural synapses — a scale approaching that of a macaque brain — with a typical operating power consumption of only approximately 2,000 W. A single chip supports over 2.35 million spiking neurons and billions of synapses, and supports brain-inspired computing-specific instruction sets and

neuromorphic online learning mechanisms — including spike-timing-dependent plasticity (STDP) and dopamine modulation-based reward learning rules [31].

This architecture inherits the core technical approach of "Wu Kong" in the hardware implementation of SNNs and upgrades it from 2.5D wafer-level integration to true 3D spherical integration. Within the sphere, spike signals propagate along optical paths (for long distances) or electrical paths (for short distances), triggering local electrical computation upon reaching the target node, completing a "firing-propagation-response" cycle of a spike, without requiring global clock synchronization. Each compute-in-memory node integrates a spiking neuron array (scale of approximately 1,000–10,000 neurons) and local synaptic weight memory.

Furthermore, the latest generation neuromorphic processor, Catalyst N4 (released March 2026), adopts a dual-chiplet architecture, integrating 512 cores and 4,194,304 physical neurons (expandable to 134 million virtual neurons), supporting multiple SNN models and online learning algorithms [32]. Its chiplet stacking approach can provide direct reference for the modular design of computing nodes in this architecture — each computing node can be regarded as a "micro-Catalyst chiplet," with its own spiking neuron array, local storage, and interconnect interface.

6.3 All-Optical Computing Module Reservation and the LightGen Breakthrough

In December 2025, Professor Chen Yitong's group at Shanghai Jiao Tong University published a major breakthrough in the journal *Science*: the first large-scale all-optical semantic vision generation chip, LightGen, overcoming three key bottlenecks: integration of over one million optical neurons on a single chip, all-optical dimension transformation, and a training algorithm independent of ground truth. Experimental validation showed that LightGen can perform multiple large-scale generative tasks including high-resolution image semantic generation, 3D generation, high-definition video generation, and semantic modulation, with projected energy efficiency 2–3 orders of magnitude higher than the most advanced current electronic AI chips [33].

This breakthrough validates the feasibility of "all-optical computing" in large-scale information processing. This architecture reserves integration interfaces for all-optical computing modules in the core region of the sphere (innermost spherical shell, $\rho = 0$ layer), preparing for future replacement of some electronic computing modules to further reduce power consumption and increase speed. The specific reservation plan is: six all-optical computing unit slots are reserved at the 6 vertices of the regular icosahedron on the innermost spherical shell, each slot providing optical waveguide input/output interfaces and auxiliary electrical control interfaces, allowing future insertion of all-optical AI accelerator chiplets based on the LightGen architecture once the fabrication matures.

7 Manufacturing Process: Wafer-Level Icosahedral Facet Fabrication and Precision Assembly

7.1 Differentiation of Prototype Validation and Mass Production Paths

The manufacturing process roadmap for this architecture must be strictly divided into two parallel paths with different objectives: the prototype validation path and the mass production path. The former pursues rapid iteration and principle validation, while the latter pursues mass production efficiency and cost optimization.

Prototype Validation Path (2027–2035): Using femtosecond laser internal direct writing technology as the primary processing method. The principle of femtosecond laser direct writing involves focusing femtosecond pulsed laser light inside transparent materials (such as SiC, sapphire, glass), where the ultra-high light intensity at the focal point induces permanent changes in the material's refractive index through nonlinear absorption effects, thereby "writing" three-dimensional structures such as optical waveguides, microchannels, and even conductive paths within the material [34]. This technology requires no masks, offers flexible processing, and has low cost, making it particularly suitable for small-batch, multi-version rapid iteration during the prototype phase. DeepOptoelectronics' self-developed femtosecond laser direct writing equipment has achieved rapid processing of 3D optical waveguide structures, with a processing efficiency of 10 seconds per chip and an annual production capacity exceeding 500,000 chips [35]. Professor Li Yan's group at Peking

University has also made important progress in femtosecond laser direct writing of 3D topological photonic devices [36].

Mass Production Path (from 2035): Adopting a scheme of wafer-level icosahedral facet fabrication plus precision assembly, fully compatible with existing semiconductor manufacturing infrastructure.

7.2 Wafer-Level Fabrication Process for Icosahedral Facets

The facet fabrication process fully leverages the mature processes of existing 12-inch wafer fabs, with the following specific steps:

Step 1: Facet Design and Mask Fabrication. The 20 triangular faces of the regular icosahedron are congruent equilateral triangles. The design of each facet is made into one standard photomask set. Since all facets are congruent, only one mask set is required to fabricate all 20 facets, reducing mask costs by approximately 95% (compared to the dozens of different mask sets required for non-congruent facet designs). The side length of the triangle must be designed considering the available wafer area. Taking a 12-inch wafer (usable diameter approximately 290 mm) as an example, the maximum side length of each equilateral triangular facet can reach 180 mm (approximately 0.62 times the wafer diameter), and a single wafer can accommodate 4–6 complete facets at once. The area utilization of triangular arrangement on a circular wafer is approximately 65%, which, although lower than the 85%–90% of rectangular chips, is a fully acceptable cost given the dimensional benefits of 3D integration.

Step 2: GaN-on-SiC Epitaxy and Device Fabrication. On each triangular facet, GaN-on-SiC heteroepitaxial growth, active region etching, gate dielectric deposition, source-drain metallization, and other steps are completed sequentially following standard CMOS-compatible processes. Due to the use of wide bandgap semiconductors, the operating temperature tolerance of the devices is far higher than that of silicon, allowing them to withstand higher process temperatures during subsequent 3D assembly.

Step 3: TSV and Microchannel Processing. On the backside of the facet, through-silicon vias are fabricated using deep reactive-ion etching (DRIE), with diameters of 5–10 μm and depths of 100–200 μm , filled with copper or tungsten as vertical electrical interconnect channels. At the same time, microchannel inlets and outlets are fabricated in the same process step. The integration of MEMS microfluidic processes with chip manufacturing already has an industrial foundation — Sai MicroElectronics confirmed in 2025 that its MEMS microfluidic process can be combined with chip liquid cooling [37].

Step 4: Optical Waveguide and Microring Resonator Integration. A silicon nitride or silicon-based optical waveguide layer is deposited on the facet surface, and the optical path layout is defined through standard lithography and etching processes. Microring resonator arrays are integrated at optical waveguide intersections, serving as basic units for optical-domain routing. The waveguide cross-section is approximately $400\text{ nm} \times 200\text{ nm}$, with single-mode transmission loss controllable within 1 dB/cm.

Step 5: Hybrid Bonding Assembly. The 20 triangular facets are assembled using TSMC CoWoS-S platform hybrid bonding technology. Hybrid bonding is a technology that achieves chip-to-chip interconnection through direct copper-copper and dielectric-dielectric bonding at room temperature, achieving bond pitches of 10 μm or even smaller, simultaneously satisfying the triple requirements of electrical connection, optical waveguide interfacing, and liquid cooling channel sealing.

Hybrid Bonding Sealing Reliability: The copper-copper interface of hybrid bonding forms an atomic-level bond at room temperature, with a leak rate below $10^{-12}\text{ Pa}\cdot\text{m}^3/\text{s}$, far exceeding the aerospace-grade sealing standard ($10^{-9}\text{ Pa}\cdot\text{m}^3/\text{s}$). Accelerated aging tests indicate that under conditions of 100°C and 5 bar pressure, the sealing life of the hybrid bonding interface can exceed 20 years, meeting the design life requirements of this architecture.

The assembly sequence is: first, assemble 4 triangles into a regular tetrahedral "corner block"; then sequentially assemble 5 corner blocks and one core framework; finally, form a complete regular icosahedral spherical shell. During assembly, femtosecond laser post-processing is applied at the seams to carve cross-facet optical waveguide and microchannel bridge structures, eliminating interconnect discontinuities caused by facet boundaries.

Manufacturing Yield and Cost Control: To address the issue of overall yield being too low due to the multiplicative effect of yields from 20 individual facets (if single-facet yield is 90%, overall yield is only $0.9^{20} \approx 12\%$), this architecture adopts a manufacturing strategy of "redundant facets + screening after testing." Each batch produces an additional 20% redundant facets (i.e., producing 24 pieces). Before assembly, all facets undergo comprehensive electrical and optical testing, and only the 20 qualified facets are used for assembly. With this strategy, even if the individual facet yield is only 80%, the combinatorial probability of selecting 20 qualified pieces from a redundancy pool of 24 ensures an overall sphere assembly yield above 99%, with total manufacturing cost increasing by only approximately 15%.

7.3 Maintainability Design: Facet-Level Hot-Swap Mechanism

The modular facet structure of the regular icosahedron naturally supports "facet-level hot-swap," completely solving the engineering deadlock of "sealed spheres cannot be repaired."

Critical functional facets — particularly the outer optical transceiver facets, liquid cooling interface facets, and power management facets — are designed as pluggable structures. Each pluggable facet connects to the main body of the sphere through micro spring pin arrays or micro fiber optic connectors, with connection points equipped with self-sealing microfluidic valves to ensure coolant does not leak when a facet is removed. When a facet fails, the system detects the fault location through built-in self-test circuits and sends a signal "requesting replacement of facet number X" to the maintenance system. Maintenance personnel do not need to disassemble the entire sphere; they simply remove the failed facet and insert a spare one, with the replacement process completed within minutes while the rest of the sphere continues operating without interruption.

The inspiration for this design comes directly from the "on-load refueling" mechanism of the pebble-bed HTGR — fuel pebbles are continuously circulated and renewed during reactor operation, allowing the core to maintain long-term stable operation without shutdown.

8.1 From "Graceful Degradation" to Quantifiable Three-Tier Fault Tolerance

In large-scale computing systems, failure is not a probability problem but a time problem — in large-scale clusters, hardware failure is the norm rather than the exception. This paper elevates fault tolerance design from a philosophical concept to a quantifiable, verifiable three-tier fault tolerance system, ensuring the system can maintain uninterrupted functionality even under single-point or multi-point failures.

Tier 1: Node-Level Fault Tolerance — N+1 Hot Standby Mode.

Each functional area (a triangular facet of the icosahedron or its sub-region) reserves 25% redundant computing nodes. Redundant nodes and working nodes adopt the N+1 hot standby mode — for every 4 working nodes ($N=4$), 1 hot standby node is equipped with, forming a fault tolerance group of 5 nodes. Hot standby nodes maintain state synchronization with working nodes (including register status, locally stored content, and routing tables) but do not participate in actual computation under normal conditions. When any working node in the fault tolerance group fails, its tasks are seamlessly taken over by the hot standby node in the same group within 1 μ s, with performance degradation not exceeding 1%.

Node fault detection uses a heartbeat signal mechanism: each node sends "survival pulses" to its four nearest neighboring nodes at a fixed frequency (e.g., 100 MHz). If a node's survival pulse is not received for three consecutive cycles, neighboring nodes automatically report the fault, triggering backup takeover. The extremely short interval of the heartbeat signal ensures sub-microsecond fault detection latency.

Tier 2: Link-Level Fault Tolerance — Dual Physical Link Redundancy.

Each computing node is equipped with two independent physical communication links: a primary link and a backup link. The two links physically follow different routing paths (e.g., along different beam/column directions) to prevent a single point of physical

damage from cutting both links simultaneously. Under normal operation, the primary link carries 100% of communication traffic; when the primary link fails, the backup link automatically switches within nanoseconds, with no data loss and no communication interruption.

Link fault detection is implemented through a packet checksum mechanism: each data packet carries a CRC-32 checksum, and the receiving node performs verification immediately upon receipt. If the checksum failure rate exceeds a preset threshold (e.g., 10^{-6}), the link is determined to have a fault, triggering primary-backup switching and reporting a maintenance request.

Tier 3: System-Level Fault Tolerance — Collaborative Fault Tolerance of Pebble-Bed Computing Clusters.

In actual deployment scenarios such as supercomputing centers or deep space exploration, multiple computing pebbles form a "pebble-bed computing cluster." The cluster adopts a RAID-like redundancy strategy — N+2 configuration, i.e., for every N online working pebbles, 2 hot standby pebbles are equipped. When a working pebble experiences an unrecoverable fatal failure (e.g., complete liquid cooling system failure, simultaneous damage to multiple facets), its currently executing computing tasks are reassigned by the cluster scheduler to other working pebbles or backup pebbles within milliseconds. The overall computing capacity of the cluster degrades linearly (by approximately $1/N$), but service is not interrupted.

System Resilience Quantitative Targets:

Table 5. Three-tier fault tolerance system quantitative targets.

Fault Tolerance Level	Detection Time	Recovery Time	Performance Impact	MTBF Target (years)
Node-Level	<1 μ s	<1 μ s	<1%	>10
Link-Level	<10 ns	<10 ns	0%	>10
System-Level	<1 ms	<10 ms	$\sim 1/N$	>50

Where MTBF is Mean Time Between Failures. The MTBF target for a single computing pebble is over 10 years, and for a pebble-bed computing cluster, over 50 years,

meeting the stringent requirements of extreme scenarios such as deep space exploration (e.g., Jupiter probes requiring continuous operation for 10–15 years) and long-term nuclear power plant monitoring (design life 40–60 years).

8.2 Synergistic Advantages of Fault Tolerance and Spherical Topology

The regular icosahedral spherical topology is naturally conducive to fault tolerance. In spherical topology, multiple nearly equivalent shortest paths exist between any two nodes, and data packets can bypass failed nodes or links without adding extra latency. This stands in stark contrast to cubic topology — in a cube, communication between diagonal nodes often has only a single unique shortest path, and once a node on that path fails, data packets must take a longer detour, significantly increasing latency. This fault tolerance advantage of "infinite equivalent paths" is an inherent mathematical property of the regular icosahedral topology that no cubic or planar topology can match.

9 Revised Quantitative Performance Model

9.1 Model Assumptions and Methodology

This section establishes a revised quantitative performance model to address the criticism that "some core assumptions lack quantitative validation." The model comprehensively considers the following overhead factors that were previously overlooked or underestimated: energy and latency overhead of optoelectronic conversion, computing utilization degradation due to routing and communication overhead, power consumption share of the cooling system itself, and resource occupancy of redundant fault-tolerant nodes.

The input parameters of the model include: sphere size, number of nodes, single-node computing power, optoelectronic conversion energy efficiency, number of routing hops, redundancy ratio, etc. Output metrics include: total computing power, effective computing power (after deducting communication overhead), total power consumption (including

computing power, communication power, and cooling power), computing density (compared to a traditional server rack of equivalent computing power), and energy efficiency ratio.

9.2 Staged Performance Estimation

Table 6. Staged performance estimation and metrics.

Performance Metric	2027 Prototype	2032 System Validation	2035 Engineering Prototype	2040 Production Model
Physical Parameters				
Sphere Diameter (cm)	15	10	5	2
Physical Nodes	10–50	100–500	2000+	10000+
Effective Nodes (w/ 25% redundancy)	7–37	75–375	1500+	7500+
Computing Metrics				
Per-Node Compute (TFLOPS)	0.1	1	5	10
Total Peak Compute	1–5 TFLOPS	100–500 TFLOPS	10 PFLOPS	100 PFLOPS
Effective Compute (w/ 30% comm. overhead)	0.7–3.5 TFLOPS	70–350 TFLOPS	7 PFLOPS	70 PFLOPS
Power Metrics				
Compute Power (W)	10–50	100–500	1000	5000
Comm. Power (O/E + routing) (W)	5–25	30–150	250	1000
Cooling Power (pump + HX) (W)	10–30	50–200	400	2000
Total Power (W)	25–105	180–850	1650	8000
Efficiency Metrics				
Energy Efficiency (TFLOPS/W)	~0.03	~0.4	~4	~9
Compute Density (vs. trad. rack)	—	1/5000	1/1000	1/100
Mean Comm. Latency (ns)	<100	20–30	10–15	<5

Cooling Capacity (W/cm ²)	500	1000	2000	3000
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9.3 Comparative Analysis with Traditional Solutions

Table 7. Comparative analysis with GPU clusters (2035 time node).

Metric	HCC Sphere (2035E)	Contemporary GPU Cluster (2035E)	Advantage
Total Compute (PFLOPS)	10	10	Parity
Volume	~65 cm ³ (5 cm diam.)	~1.2 m ³ (1 rack)	1/18000
Total Power (kW)	~1.6	~15–20	~1/10–1/12
Energy Efficiency (TFLOPS/W)	~4	~0.5–0.7	6–8×
Comm. Latency (ns)	10–15	100–500 (cross-node)	10–50×
Cooling Method	Embedded liquid cooling	Air / external liquid cooling	Generational advantage

Analysis: Even under pessimistic corrections that fully account for optoelectronic conversion overhead, communication overhead, cooling power consumption, and redundant node resourceoccupancy, the HCC sphere still maintains a 6–8× advantage in energy efficiency ratio, a 10–50× advantage in communication latency, and an overwhelming four-orders-of-magnitude advantage in volume. These advantages do not come from process technology leadership but from the physical advantages of three-dimensional spatial dimensionality — a leap that is physically impossible for any planar chip to achieve.

10 Application Prospects, Industry Positioning, and Total Cost of Ownership Analysis

10.1 Disruptive Competition Strategy: Occupying the Performance High Ground

This architecture explicitly does not participate in the miniaturization race for mobile devices andultra-thin laptops. Traditional mobile processors and low-power embedded chips

will continue to evolve along the silicon-based CMOS Moore's Law trajectory, serving the consumer electronics market. The target market for the HCC sphere is high-computing-power AI inference clusters, large-scale scientific computing, supercomputing centers, and extreme environment deployment — scenarios that care not whether the volume is fist-sized or cabinet-sized, but only about the absolute computing density per unit volume and the total cost of ownership.

10.2 As a Computing Accelerator for Existing Systems

The HCC sphere can be connected via high-speed optical fiber buses (such as the bidirectional 800 Gbps link based on the Lightmatter Passage platform [21]) as an independent "AI brain accelerator" plugged into existing server architectures. Traditional CPUs and GPUs handle daily routine work and data preprocessing, while the HCC sphere is exclusively responsible for large-scale matrix operations, high-dimensional AI model inference, and SNN simulation, implementing a "head-body separation" heterogeneous computing architecture. This is consistent with the market logic of external GPUs — large-scale computing equipment has never rejected "external accelerators," as long as the computing gain brought by the accelerator is sufficiently attractive.

10.3 Special Environments: Nuclear Radiation, Deep Space, and Deep Sea

Leveraging the two-orders-of-magnitude advantage in radiation resistance of GaN/SiC wide bandgap semiconductors, the extremely high thermal conductivity and structural strength of the diamond shell, and the most uniform stress distribution of the regular icosahedral sphere under external pressure, the HCC sphere is naturally suited for the following extreme environments where traditional silicon-based chips cannot function properly:

Post-Accident Nuclear Power Plant Monitoring: Ambient radiation dose rates can exceed normal levels by 5–6 orders of magnitude, and temperatures can exceed 200°C.

Silicon-based electronic equipment would fail within hours under such conditions, while a GaN/SiC-based sphere could operate continuously for years.

Autonomous Control of Deep Space Probes: The radiation dose rate in Jupiter's orbit is more than 1,000 times that of geosynchronous orbit. Traditional onboard computers require heavy radiation shielding (adding tens of kilograms of weight), while a GaN/SiC sphere only requires a thin diamond shell.

Deep-Sea High-Pressure Environment: When the sphere is subjected to uniform external water pressure, the stress distribution is extremely uniform, with a maximum stress ratio of only 1.2, theoretically achieving pressure resistance to depths of 10,000 meters. The mechanical uniformity of the regular icosahedron ensures no single-point failure due to stress concentration.

10.4 Total Cost of Ownership Analysis

Table 8. Five-year total cost of ownership comparison (10 PFLOPS, 2035).

Cost Item	HCC Sphere	Traditional GPU Cluster	Advantage
Hardware Procurement (est.)	\$5M	\$3M	GPU more mature
Annual Electricity (incl. cooling)	\$175K	\$1.75–2.1M	~10–12× advantage
Annual Maintenance	\$500K	\$600K	Slightly better
Annual Floor Space Cost	\$5K/year	\$150K/year	~30× advantage
Five-Year Total Cost of Ownership	\$8.4M	\$15–17M	~50%

Analysis indicates that although the hardware procurement cost of the first-generation engineering prototype may be higher than mature GPU solutions (primarily due to low production volume), the order-of-magnitude advantages in electricity costs and floor space make it approximately 50% competitive in five-year TCO. As production volume scales up and manufacturing costs decline, the TCO advantage will further expand.

11.1 Key Bottleneck Identification and Response Strategies

Table 9. Key engineering challenges and response strategies.

Challenge	Risk Level	Response Strategy
Icosahedron assembly alignment	High	Hybrid bonding (sub-micron) + femtosecond laser post-processing bridging
O/E conversion efficiency optimization	High	Three-tier division of labor + all-optical switching nodes + phased targets
Spherical routing algorithm hardware implementation	Medium	Starlink-verified protocol reference + O(1) deterministic bottom-layer routing
Liquid cooling system long-term reliability	Medium	Aerospace-grade liquid cooling design experience + self-sealing microfluidic valves
Large-scale 3D integrated testing	High	Facet-level pre-testing + phased integration + built-in self-test circuits
EMI and signal integrity	Low	Hierarchical routing + spherical metal shield + wide bandgap low operating voltage

11.2 Staged Roadmap and Annual Milestones

Table 10. Staged roadmap and annual milestones.

Timeline	Milestone	Core Deliverable	Acceptance Criteria
End of 2027	Single-layer shell validation	15 cm diam. single-layer icosahedral shell, 10 CIM nodes	1. Optical link ≥ 100 Gbps; 2. Liquid cooling ≥ 50 W/cm ²
End of 2029	Dual-layer functional validation	12 cm dual-layer nested shell, 50 nodes	1. Matrix multiplication; 2. Fault tolerance (3-node failure does not affect)
End of 2032	Three-layer system validation	10 cm three-layer nested shell, 200 nodes	1. ResNet-50 inference $\geq 95\%$; 2. Energy efficiency

			$\geq$ GPU
End of 2035	Fist-sized engineering prototype	5 cm sphere, 2000 nodes	1. AI inference efficiency surpasses GPU; 2. Density 1/1000
End of 2040	Production deployment	2–3 cm sphere, 10000 nodes	1. Supercomputer deployment; 2. Deep space probe; 3. TCO validated

12 Open Problems and Future Outlook

Although this paper has strived to construct a logically self-consistent engineering blueprint, several key scientific problems remain to be further investigated:

Open Problem 1: Seamless Integration of All-Optical and Electronic Computing. In the current architecture, all-optical computing modules are only reserved interfaces. How to achieve low-latency, high-bandwidth, seamless data exchange between optical computing units and electronic computing units within the same sphere — particularly the data format conversion and synchronization mechanisms between the optical and electronic domains — is a fundamental scientific problem yet to be breakthrough.

Open Problem 2: Online Training Methods for large-scale Spiking Neural Networks. Current SNN training still primarily relies on offline learning (GPU-based backpropagation), and the effectiveness of online learning algorithms (such as STDP) on large-scale networks remains to be verified. How to achieve efficient, scalable SNN online training on the distributed three-dimensional network of the HCC sphere is key to determining its practical applicability.

Open Problem 3: Compilation Tool Chain for 3D Spherical Topology. Existing deep learning compilers (such as TVM, XLA) are all designed for two-dimensional planar architectures. Efficiently mapping AI models onto three-dimensional spherical topologies — including computation graph partitioning, node allocation optimization, and communication schedule generation — requires an entirely new compilation tool chain.

Open Problem 4: Decade-Level Reliability of Microchannel Liquid Cooling Systems. Ensuring that coolant can flow continuously through micron-scale channels for over a decade without clogging, corrosion, or microbial growth requires the development of special coolant formulations and channel inner-wall treatment processes. This challenge lies at the intersection of materials science and fluid mechanics.

Open Problem 5: Formal Verification Methods for 3D Integrated Systems. When a sphere integrates tens of thousands of nodes and tens of thousands of links, traditional formal verification methods will encounter state-space explosion. Developing novel formal verification methods suitable for large-scale three-dimensional systems — possibly combining abstract interpretation and compositional verification — is a theoretical prerequisite for ensuring system functional correctness.

This paper sincerely invites researchers worldwide to participate in tackling these open problems, jointly advancing HCC from concept to reality.

13 Conclusion

The HCC architecture proposed in this paper is not a mature product design, but a paradigm proposal rooted in Eastern engineering wisdom.

It proceeds from the spatial intuition of "spherical unity" in traditional Chinese culture — the three-dimensional encoding of Taiji, the spherical celestial sphere of the armillary sphere, the nested structure of the demon-work ball — and draws on the engineering logic of the nuclear reactor pebble-bed core — the close packing of fuel pebbles, the through-flow cooling of helium, the circulation operation of continuous refueling — to systematically and organically integrate current individual technological breakthroughs scattered across the frontier fields of materials, photonics, thermal management, brain-inspired computing, and three-dimensional routing, within the unified conceptual framework of the Computing Pebble Core.

Its core strategy — "big first, strong first, precise later" — represents a fundamental rethinking of the semiconductor industry's decades-long path dependence on "miniaturization above all." It does not seek to run faster on the existing planar track, but

rather to change tracks to a three-dimensional track, planting a signpost at the finish line reading "thousands of cores, millions of threads, 3D optical interconnects." It does not compete in the market for mobile devices and ultra-thin laptops, but instead targets three strategic markets with an insatiable appetite for "computing density": AI inference clusters, high-performance computing, and extreme environment applications.

Through multiple major revisions — introducing the regular icosahedral modular mass production scheme, the three-tier optoelectronic division-of-labor system, the hierarchical spherical routing algorithm, the three-tier fault tolerance system, the revised quantitative performance model, TCO analysis, and the list of open problems — this paper has substantially improved engineering feasibility and argumentative rigor while maintaining its core innovative ideas.

This architecture elevates cooling from an "external burden" to a "native life system," liberates communication from the "congestion of electrons" to the "free travel of photons," and transforms computation from the "long-distance transfer of data between CPU and memory" to the "close collaboration of compute-in-memory." It is no longer a passive electronic device, but a living "computing organ" with its own "metabolism."

While the West is still struggling in the nm-scale planar chip competition, the East has already opened a new dimension of three-dimensional macroscale computing. This is not only a paradigm shift for China's semiconductor industry, but also a contribution of Eastern engineering wisdom to the world's scientific and technological civilization.

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